Literature Survey on PIR-Based Private Video Retrieval

CS670 - Cryptographic Protocols & Techniques

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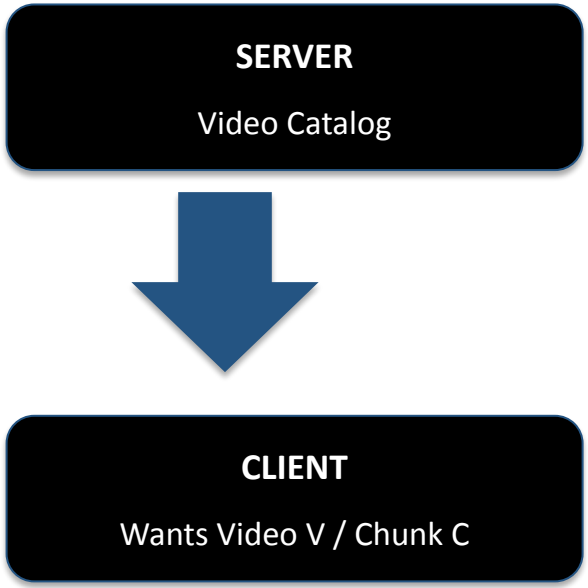
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Project Problem & Motivation

Conventional video streaming exposes the requested content to the service provider.

- The server can learn which video is being accessed.
- The server can learn which video chunk is being requested.
- Repeated requests can reveal the client's viewing pattern.

Goal: retrieve video content without revealing what the client is watching.



Project Objective

Design and implement a PIR-based video retrieval system that enables private retrieval of video chunks while sustaining real-time playback.

PRIVACY

Hide which video and chunk the client retrieves.

REAL-TIME

Each batch must fit within the playback buffer's time budget.

SCALABILITY

Evaluate across realistic catalog sizes.

NETWORK

Evaluate under different network conditions.

Privacy is not enough — the system must also be fast enough for continuous playback.

What is Private Information Retrieval?

Client wants to retrieve an item from a server-held database.

Privacy requirement: the server should not learn which database item the client requested.

Let $D = (D_1, D_2, \ldots, D_n)$ be the database and D_i the desired item.

PIR provides a mechanism for obtaining D_i while hiding the index i from the server.



Types of PIR

Information-Theoretic PIR (IT-PIR)

- Privacy does not rely on computational hardness assumptions.
- Typically uses multiple non-colluding servers.
- Provides information-theoretic privacy.

Computational PIR

- Privacy relies on computational assumptions.
- Can be deployed in the single-server setting.
- Cryptographic computation introduces efficiency costs.

Why Practical PIR?

COMMUNICATION

Large queries or responses can dominate network cost.

COMPUTATION

Cryptographic processing can be expensive for large databases.

ONLINE LATENCY

Online work directly affects retrieval time.

SCALABILITY

Costs must remain manageable as the catalog grows.

These bottlenecks motivated a progression of increasingly efficient PIR systems.

Evolution of Practical PIR



Research progressively reduces communication and computation overhead to make single-server PIR more practical.

XPIR: Practical Single-Server PIR

Goal: demonstrate a practical computational PIR system in the single-server setting.

Approach: lattice-based homomorphic encryption lets the server process a concealed query.

Privacy: the server performs retrieval computation without learning the requested database index.

Significance: an important practical baseline for subsequent single-server PIR research.

Limitation: homomorphic computation can introduce substantial computation and communication overhead.

XPIR establishes practicality, but efficiency remains a central challenge.

Communication-Efficient PIR

SealPIR

Query-side efficiency

- Compressed queries
- Amortized processing
- Reduced query communication

OnionPIR

Response-side efficiency

- Reduces PIR response size
- Targets communication overhead
- Trades communication against cryptographic computation

SealPIR primarily targets query overhead; OnionPIR targets response overhead.

Computationally Efficient PIR

Work	Main direction
Sublinear Online Time	Reduce computation required during the online phase
SimplePIR	Efficient single-server PIR with reduced amortized server computation
SPIRAL	High-performance PIR through efficient homomorphic computation
FrodoPIR	Efficient LWE-based PIR approach
Piano	Sublinear online computation and scalability

Key Takeaways from Practical PIR

Communication

Reduce query and response overhead

Computation

Reduce server-side cryptographic work

Online latency

Reduce or move work in the online phase

Scalability

Make costs manageable for large databases

But video retrieval requires multiple upcoming chunks to be retrieved efficiently and on time.

→ This motivates Batch PIR.

Why Batch PIR for Video?

Vectorized BatchPIR

From PIR to Private Streaming

Private Streaming with Convolutional Codes

Popcorn: Private Media Consumption

What Does Existing Streaming Work Achieve?

PIR in Real-Time Applications

METADATA-PRIVATE MESSAGING

Addra hides voice-call metadata via PIR-retrieved private mailboxes; message latency is directly bounded by PIR retrieval time.

PRIVATE MEDIA STREAMING

Popcorn applies PIR to a Netflix-scale video catalog, batching concurrent users' requests to amortize server cost.

PRIVATE CONTACT DISCOVERY

Messaging apps must privately resolve large contact databases within a sub-second, user-facing latency budget.

LOW-LATENCY LOOKUPS

Private DNS- and search-style queries are expected to match the response time of their non-private counterparts.

Across every application, PIR's real-world viability is decided by whether server computation and communication fit the latency budget — not by cryptographic security alone.

Addra / FastPIR

Problem: Existing PIR schemes can introduce heavy server computation and communication overhead, hindering latency-sensitive applications.

System: Addra hides voice-call metadata by having callers and callees exchange messages through PIR-retrieved private mailboxes on fully untrusted infrastructure.

FastPIR: A new computational PIR scheme built to reduce server-side answer-processing latency, avoiding recursion to keep the response size small.

Result: Serves 32,000 users with 726ms 99th-percentile message latency on an 80-machine AWS cluster — a 7x improvement over the prior text-messaging system in the same threat model.

Communication cost: About 1.46 Mbps download bandwidth per user; 96-byte voice messages expand to roughly 64KB ciphertexts (≈682x).

Relevance: Demonstrates PIR must be evaluated end-to-end — latency and bandwidth per user — not by cryptographic operation counts alone.

Addra shows real-time-capable PIR is achievable — but requires an application-specific scheme and heavy infrastructure parallelism.

What Should We Compare?

To select a design for private, real-time video retrieval, we compare surveyed systems along six axes:

COMMUNICATION COST

Query + response size transferred per retrieval.

COMPUTATION COST

Server-side (and client) processing time per query.

BATCH SUPPORT

Whether the scheme retrieves several chunks more cheaply than one-by-one.

ONLINE/OFFLINE SPLIT

How much per-query work can be precomputed before the request arrives.

THREAT MODEL

Information-theoretic multi-server vs. computational single-server trust assumptions.

REAL-TIME FIT

Whether online latency fits inside the playback buffer's time budget.

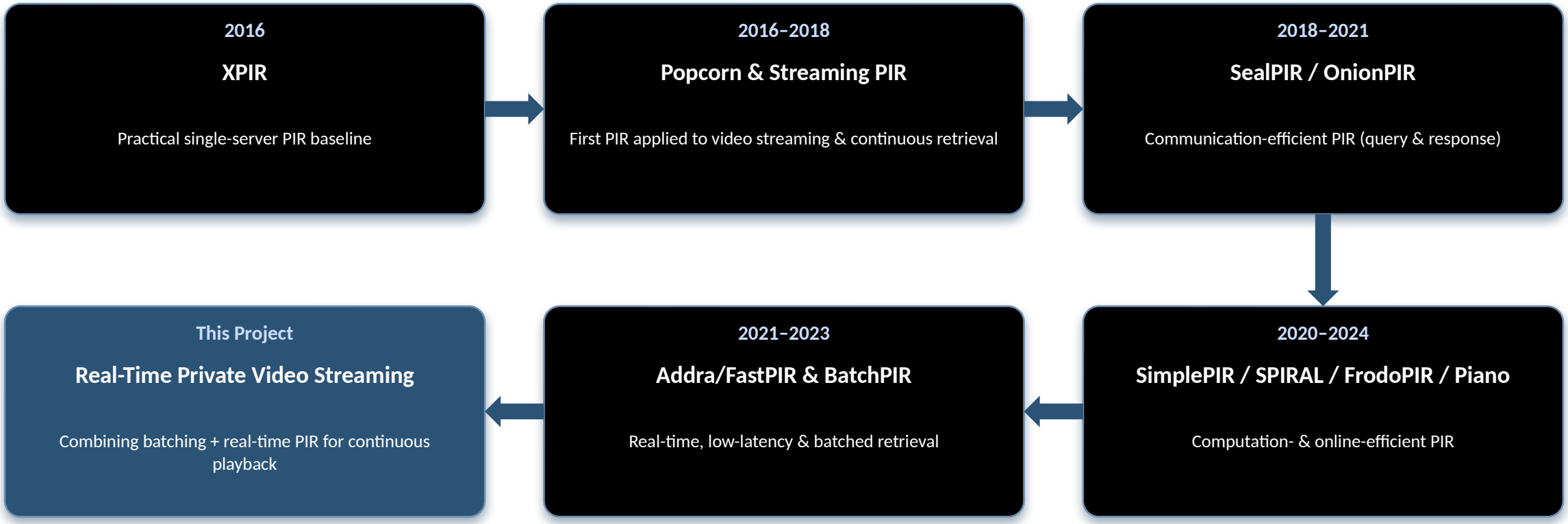
These six axes define the comparison table and evolution timeline that follow.

Literature Comparison

System	Type	Comm.	Compute	Batch	Real-Time Fit
XPIR	Computational	Low	Low	No	Poor
SealPIR	Computational	High	Medium	Partial (PBC)	Moderate
OnionPIR	Computational	High	Medium	No	Moderate
SimplePIR	Computational	Medium	High (amortized)	No	Moderate
SPIRAL	Computational	High	High	No	Good
FrodoPIR	Computational	Medium	High (offline-heavy)	No	Good
Piano	Computational	Medium	Very High (sublinear)	No	Good
Vectorized BatchPIR	Computational	High	High	Yes (native)	Good
Popcorn (system)	Computational	Medium	Medium (batched)	Yes (cross-user)	Moderate
Addra / FastPIR (system)	Computational	Low (per-msg)	High	Implicit (mailbox rounds)	Good

Ratings are qualitative, drawn from each paper's reported benchmarks under differing setups — not directly comparable numeric figures.

Evolution of the Literature



Fifteen years of PIR research have produced the individual building blocks — real-time, batched, private video retrieval is the open combination this project targets.

Research Gap

Motivation for Our Project

Proposed Evaluation / System Perspective

Conclusion